KS3 DIGITAL TECHNOLOGY

DIGITAL MONEY AND FINTECH

How money became data: how card and phone payments work behind the scenes, how app banks and open banking use technology, how payments are secured, how to spot scams and avoid becoming a money mule, what cryptocurrency and blockchain really are, and how AI and data are shaping the future of money.

MR BUTLER'S DIGITAL TECHNOLOGY

LESSONS

1 Tap to pay
2 Fintech and app banking
3 Securing payments
4 Scams and money mules
5 Crypto and blockchain
6 The future of money

LESSON 1 OF 6

TAP TO PAY

When you pay by card or phone, a chain of computer systems checks the payment in about a second: the terminal, the shop's bank, the card network and your bank.

DIGITAL MONEY AND FINTECH

STARTER · 5 MINUTES

DO NOW

When did you last pay with cash? How did you pay for the last thing you bought?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how card, contactless and mobile payments work and the systems involved

SUCCESS CRITERIA: I CAN...

- ✓ I can describe how payments have changed
- ✓ I can describe the steps in a card payment
- ✓ I can explain how contactless and phone payments work

MONEY IS DATA

- Most money exists only as numbers in bank computer systems
- Cash is now used for only a small share of UK payments
- Contactless cards, phones and watches are the most common ways to pay in shops
- Bank transfers are almost instant with Faster Payments

KEY WORDS

CONTACTLESS	A way to pay by tapping a card or device on a reader.
NFC	Near-field communication: short-range radio used for contactless payments.
CARD NETWORK	A company such as Visa or Mastercard that connects banks to process card payments.
AUTHORISATION	A bank approving or declining a payment.
DIGITAL WALLET	An app that stores payment cards on a phone or watch, such as Apple Pay or Google Pay.

WHAT HAPPENS WHEN YOU TAP

- Terminal reads the card or phone using NFC
- Request goes to the shop's bank, then the card network (Visa, Mastercard)
- Your bank checks: enough money? Looks like fraud?
- Approved or declined, all in about a second

ACTIVITY · 20 MINUTES · GROUPS OF 5

PAYMENT RELAY

1. Each person takes a role: customer, terminal, shop's bank, card network, customer's bank.

2. Pass a "payment message" card along the chain, adding the information each role checks.

3. Run three scenarios: a normal payment, not enough money, and a suspicious payment from another country.

4. Draw a diagram of the payment chain.

In your workbook, answer:

? What does the customer's bank check before approving a payment?

? What happened in the suspicious payment scenario?

Your workbook: Lesson 1 · tick each step off as you go

PHONE WALLETS

- Apple Pay and Google Pay store a token, not your real card number
- You unlock with fingerprint, face or PIN
- Each payment uses a one-time code
- Safer than a physical card if your phone is stolen

QUICK CHECK

1. What technology does contactless payment use?

A NFC (near-field communication)
B Bluetooth only
C Satellites
D Cables

2. Which is a card network?

A Monzo
B Amazon
C Visa
D PayPal

3. What does a digital wallet store instead of your real card number?

A Your PIN
B Your password
C Your address
D A token

4. What is authorisation?

A Your bank approving a payment
B Printing a receipt
C Writing a book
D Moving money the next day

QUICK CHECK: ANSWERS

1. **A: NFC (near-field communication)**
Short-range radio.

2. **C: Visa**
Visa and Mastercard connect banks.

3. **D: A token**
Tokens protect card details.

4. **A: Your bank approving a payment**
Approve or decline.

4 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe the steps when a customer pays by contactless card.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe how payments have changed
- ✓ I can describe the steps in a card payment
- ✓ I can explain how contactless and phone payments work

LESSON 2 OF 6

FINTECH AND APP BANKING

Fintech (financial technology) uses software to provide financial services.

STARTER · 5 MINUTES

DO NOW

What can you do on a banking app that you couldn't do in a bank branch?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain what fintech is and how app-based banks, open banking and APIs are changing financial services

SUCCESS CRITERIA: I CAN...

- ✓ I can explain what fintech is and give examples
- ✓ I can explain open banking and the role of APIs
- ✓ I can evaluate fintech products such as Buy Now, Pay Later

WHAT IS FINTECH?

- Fintech = financial technology
- App-based banks, payment apps, budgeting apps, money transfer apps
- The UK, and London in particular, is one of the world's biggest fintech hubs
- Northern Ireland has a growing fintech and financial software industry

KEY WORDS

FINTECH	Financial technology: using software and technology to provide financial services.
OPEN BANKING	A system letting customers securely share their bank data with approved apps, with consent.
API	Application programming interface: rules that let one program request data or services from another.
BUY NOW PAY LATER	A way of splitting a purchase into several payments; a form of borrowing.
REGULATION	Rules set and enforced by an official body.

OPEN BANKING AND APIS

- An API lets one program request data or services from another
- Open banking: you can let approved apps access your bank data securely
- e.g. a budgeting app shows all your accounts in one place
- You give consent and can withdraw it at any time

BUY NOW, PAY LATER?

- BNPL lets you split payments, often with no interest if paid on time
- Easy to use: one click at checkout
- Risk: building up debt across several purchases
- Missed payments can lead to fees and harm your credit record

ACTIVITY · 20 MINUTES · PAIRS

FINTECH REVIEW

1. Look at the screenshots of three (fictional) fintech apps: a teen banking app, a budgeting app using open banking, and a BNPL checkout.
2. For each, list two features that help users and one risk.
3. Decide which app you'd recommend for a 14-year-old and explain why.
4. Write one question you'd ask before connecting your bank account to an app.

In your workbook, answer:

? Which app would you recommend for a 14-year-old, and why?

? What should you ask before connecting your bank account to an app?

Your workbook: Lesson 2 · tick each step off as you go

QUICK CHECK

1. What does fintech stand for?

A Finished tech
B Final technique
C Fine technology
D Financial technology

2. What is an API?

A Rules that let one program request data or services from another
B A type of bank
C A password
D A coin

3. What does open banking allow?

A Free money
B Banks to close
C Sharing your bank data securely with approved apps, with your consent
D Anyone to see your account

4. Which organisation regulates financial firms in the UK?

A Ofcom
B The DVLA
C The Financial Conduct Authority
D The CAA

QUICK CHECK: ANSWERS

1. **D: Financial technology**
   Tech for money.

2. **A: Rules that let one program request data or services from another**
   Programs talking to programs.

3. **C: Sharing your bank data securely with approved apps, with your consent**
   Consent is key.

4. **C: The Financial Conduct Authority**
   FCA.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain what open banking is.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain what fintech is and give examples
- ✓ I can explain open banking and the role of APIs
- ✓ I can evaluate fintech products such as Buy Now, Pay Later

LESSON 3 OF 6

SECURING PAYMENTS

Digital payments are protected by several layers: encryption scrambles data in transit, tokenisation replaces card numbers, strong customer authentication requires two different types of proof, and AI systems analyse every payment in real time to spot fraud.

DO NOW

What stops someone who finds your card number online from spending your money?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain how encryption, tokenisation, strong authentication and AI fraud detection protect digital payments

SUCCESS CRITERIA: I CAN...

- ✓ I can explain how encryption and tokenisation protect payment data
- ✓ I can explain strong customer authentication
- ✓ I can describe how AI detects fraud

SCRAMBLE AND SUBSTITUTE

- Encryption: scrambles data so only the right system can read it
- HTTPS and the padlock protect card details sent to websites
- Tokenisation: a substitute number replaces the real card number
- A stolen token is useless elsewhere

KEY WORDS

ENCRYPTION	Scrambling data so only someone with the right key can read it.
TOKENISATION	Replacing a real card number with a substitute number.
STRONG CUSTOMER AUTHENTICATION	Proving who you are with two different types of evidence.
BIOMETRICS	Using body features such as fingerprints or faces to prove identity.
FRAUD DETECTION	Systems, often using AI, that spot and stop fraudulent payments.

PROVE IT'S YOU

- Strong customer authentication needs two of three types of proof
- Something you know: PIN or password
- Something you have: phone or card
- Something you are: fingerprint or face (biometrics)

AI ON GUARD

- AI checks every payment in milliseconds
- Looks for unusual amounts, places, times or devices
- Can block payments or ask you to confirm in your app
- False positives: sometimes genuine payments get blocked

ACTIVITY · 20 MINUTES · PAIRS

BE THE FRAUD AI

1. Read the spending profile of "Erin", 15, who lives in Enniskillen.
2. For 12 new payments, decide: approve, ask Erin to confirm, or block.
3. Write three rules your fraud AI follows.
4. Compare with another pair: where did you disagree and why?

In your workbook, answer:

? Write your three fraud rules.

? Give an example where your rules would make a false positive.

Your workbook: Lesson 3 · tick each step off as you go

QUICK CHECK

1. What does encryption do?

A Scrambles data so only the right system can read it
B Speeds up payments
C Prints receipts
D Deletes data

2. What is tokenisation?

A A type of password
B Collecting tokens at an arcade
C Replacing a card number with a substitute number
D Printing coins

3. Which pair counts as strong customer authentication?

A A password and a code sent to your phone
B Your name and address
C A password and a memorable word
D Two passwords

4. What might AI fraud detection do with an unusual payment?

A Nothing
B Approve it automatically
C Delete your account
D Block it or ask you to confirm

QUICK CHECK: ANSWERS

1. **A: Scrambles data so only the right system can read it**
   Protects data in transit.

2. **C: Replacing a card number with a substitute number**
   Tokens are useless if stolen.

3. **A: A password and a code sent to your phone**
   Two different types.

4. **D: Block it or ask you to confirm**
   It spots unusual patterns.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain strong customer authentication, with an example.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain how encryption and tokenisation protect payment data
- ✓ I can explain strong customer authentication
- ✓ I can describe how AI detects fraud

LESSON 4 OF 6

SCAMS AND MONEY MULES

Criminals use phishing, fake websites, impersonation and social media to trick people into sending money or details.

DO NOW

You see a post: "Earn £500 this week! Just let us use your bank account." What questions should you ask?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Recognise common payment scams, explain why money muling is a crime, and describe how to protect yourself

SUCCESS CRITERIA: I CAN...

- ✓ I can recognise common scam techniques
- ✓ I can explain what a money mule is and the consequences
- ✓ I can describe how to protect myself and where to get help

SCAM TRICKS

- Phishing (email), smishing (text) and fake websites
- Impersonation: pretending to be your bank, the police or a delivery firm
- Urgency and pressure: "act now or lose your money"
- Too good to be true: fake prizes, investments, cheap tickets

KEY WORDS

PHISHING	Fake messages designed to trick people into giving away details or money.
AUTHORISED PUSH PAYMENT FRAUD	When someone is tricked into sending money to a criminal themselves.
MONEY MULE	Someone who lets criminals move money through their bank account, which is a crime.
SOCIAL ENGINEERING	Manipulating people into giving away information or money.

MONEY MULES

- Criminals ask young people to receive and pass on money "for a fee"
- Advertised as easy cash, jobs or "favours" on social media
- The money comes from crime such as fraud or drugs
- It's money laundering: accounts closed, possible prison, hard to get a phone contract or student loan

STOP. CHALLENGE. PROTECT.

- Stop: take a moment before sharing details or sending money
- Challenge: could it be fake? It's OK to say no
- Protect: contact your bank directly; tell a trusted adult
- Call 159 to reach your bank safely if you think you've been scammed

ACTIVITY · 25 MINUTES · PAIRS

SCAM SPOTTER

1. Look at the eight (fictional) messages and adverts, including a parcel text, a "job" advert on social media and a cheap concert ticket.
2. Decide which are scams and highlight the warning signs.
3. For the "job" advert, explain why it's money muling and what could happen to someone who agrees.
4. Write a short warning post for younger pupils.

In your workbook, answer:

? What warning signs are in the "job" advert?

? What should you do if you've already shared your bank details?

Your workbook: Lesson 4 · tick each step off as you go

QUICK CHECK

1. What is a money mule?

A A savings account
B Someone who lets criminals move money through their account
C A farm animal
D A bank worker

2. Will your bank ever ask you to move money to a "safe account"?

A Yes, often
B No, never
C Only by text
D Only on Fridays

3. What number can you call to reach your bank safely in the UK?

A 159
B 999
C 111
D 101

4. What is social engineering?

A Building bridges
B A type of encryption
C Manipulating people into giving information or money
D A social media app

QUICK CHECK: ANSWERS

1. **B: Someone who lets criminals move money through their account**
It's a crime.

2. **B: No, never**
It's always a scam.

3. **A: 159**
A secure service.

4. **C: Manipulating people into giving information or money**
Hacking humans.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe three warning signs of a payment scam.

Look back at today's success criteria: can you tick them all?

- ✓ I can recognise common scam techniques
- ✓ I can explain what a money mule is and the consequences
- ✓ I can describe how to protect myself and where to get help

LESSON 5 OF 6

CRYPTO AND BLOCKCHAIN

A blockchain is a shared, tamper-resistant digital record (ledger) stored on many computers.

STARTER · 5 MINUTES

DO NOW

If there's no bank, who keeps track of who owns what money?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain what blockchain and cryptocurrency are, and evaluate their benefits and risks

SUCCESS CRITERIA: I CAN...

- ✓ I can explain what a blockchain is
- ✓ I can explain how cryptocurrency works and how it differs from normal money
- ✓ I can evaluate the risks of cryptocurrency

A SHARED RECORD

- A ledger is a record of transactions
- A blockchain is a ledger shared across many computers
- Transactions are grouped into blocks, linked using cryptographic hashes
- Changing an old block would break the chain, so tampering is easy to spot

KEY WORDS

BLOCKCHAIN	A shared, tamper-resistant digital ledger stored on many computers.
LEDGER	A record of financial transactions.
CRYPTOCURRENCY	Digital money recorded on a blockchain rather than by a bank.
VOLATILITY	How much and how quickly a price goes up and down.
STABLECOIN	A cryptocurrency linked to normal money to keep a steady value.

CRYPTOCURRENCY

- Digital currency recorded on a blockchain, with no central bank
- Bitcoin (2009) was the first; thousands now exist
- New blocks are added by mining (proof of work) or staking (proof of stake)
- Stablecoins are linked to normal money like the US dollar

ACTIVITY · 25 MINUTES · GROUPS OF 4

BUILD A PAPER BLOCKCHAIN

1. Each group starts a ledger of class "tokens". Each block (an index card) records 3 transactions and the "fingerprint" of the previous block (use the simple hash your teacher gives you, e.g. add up all the numbers in the block).
2. Every group keeps a copy of every block.
3. One pupil secretly changes a transaction in an early block. Can the class find it?
4. Discuss: what are the strengths and weaknesses of this system compared with a bank?

In your workbook, answer:

? How did the class spot the changed transaction?

? Give one strength and one weakness of the blockchain compared with a bank.

Your workbook: Lesson 5 · tick each step off as you go

RISKS

- Very volatile: prices can halve in weeks
- Not protected like bank savings; lost keys mean lost money
- Widely used in scams, hacks and crime
- Proof-of-work mining uses huge amounts of electricity

QUICK CHECK

1. What is a blockchain?

A A bank vault
B A type of password
C A chain made of blocks
D A shared, tamper-resistant digital ledger stored on many computers

2. What was the first cryptocurrency?

A Dogecoin
B Bitcoin
C Ethereum
D The digital pound

3. What does volatile mean for a cryptocurrency?

A It explodes
B It's stable
C It's illegal
D Its price can change a lot very quickly

4. What is a stablecoin?

A A coin from a stable
B A cryptocurrency linked to normal money to keep a steady value
C A type of bank card
D A heavy coin

QUICK CHECK: ANSWERS

1. **D: A shared, tamper-resistant digital ledger stored on many computers**
Linked blocks of transactions.

2. **B: Bitcoin**
Launched in 2009.

3. **D: Its price can change a lot very quickly**
Big price swings.

4. **B: A cryptocurrency linked to normal money to keep a steady value**
e.g. linked to the US dollar.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why a blockchain is described as tamper-resistant.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain what a blockchain is
- ✓ I can explain how cryptocurrency works and how it differs from normal money
- ✓ I can evaluate the risks of cryptocurrency

LESSON 6 OF 6

THE FUTURE OF MONEY

AI is used in finance for credit decisions, fraud detection, customer chatbots and trading.

DO NOW

Should a computer decide whether someone gets a loan? What information should it be allowed to use?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Evaluate the use of AI and data in finance and the move towards a cashless society, and design a responsible fintech solution

SUCCESS CRITERIA: I CAN...

- ✓ I can describe how AI and data are used in finance
- ✓ I can discuss the impacts of a cashless society
- ✓ I can design a responsible fintech app

AI AND YOUR MONEY

- Credit scoring: AI helps decide who gets loans and phones on contract
- Fraud detection and security checks
- Chatbots answer customer questions
- AI trading systems buy and sell shares in milliseconds

KEY WORDS

CREDIT SCORE	A number lenders use to judge how likely someone is to repay borrowing.
ALGORITHMIC BIAS	When an algorithm or AI treats some groups unfairly.
CASHLESS SOCIETY	A society where all payments are made digitally.
FINANCIAL INCLUSION	Everyone having access to useful, affordable financial services.

FAIR OR BIASED?

- AI learns from past data, which may reflect past unfairness
- Algorithmic bias could treat some groups unfairly
- People have a right to an explanation of important automated decisions
- Regulators expect firms to check AI for fairness

GOING CASHLESS?

- Card and phone payments are fast and convenient
- But some people rely on cash: older people, people in rural areas, people budgeting carefully
- Cash still works in a power cut or system outage
- UK rules now protect access to cash in local communities

ACTIVITY · 35 MINUTES · GROUPS OF 3

DESIGN A MONEY APP

1. Design a fintech app for 11-16 year olds that helps them manage money safely.
2. Plan three features (e.g. savings goals, spending insights, scam alerts) and sketch the main screen.
3. Explain how your app keeps users secure and protects their privacy.
4. Explain how your app avoids encouraging overspending, and pitch it to the class.

In your workbook, answer:

? Describe your three features and how they help.

? How does your app protect users' security and wellbeing?

Your workbook: Lesson 6 · tick each step off as you go

QUICK CHECK

1. What is a credit score?

A A test score
B A number lenders use to judge how likely you are to repay
C A football result
D A bank balance

2. What is algorithmic bias?

A A type of encryption
B When an AI treats some groups unfairly
C A secure system
D A fast algorithm

3. Which is an argument against a cashless society?

A Easier tax collection
B Less cash crime
C Faster payments
D It excludes people without bank accounts or smartphones

4. What is financial inclusion?

A Including finance in maths lessons
B Only rich people banking
C Everyone having access to useful, affordable financial services
D Closing banks

QUICK CHECK: ANSWERS

1. **B: A number lenders use to judge how likely you are to repay**
   Based on borrowing history.

2. **B: When an AI treats some groups unfairly**
   Often learned from past data.

3. **D: It excludes people without bank accounts or smartphones**
   Inclusion matters.

4. **C: Everyone having access to useful, affordable financial services**
   Access for all.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain how algorithmic bias could affect decisions about loans.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe how AI and data are used in finance
- ✓ I can discuss the impacts of a cashless society
- ✓ I can design a responsible fintech app

END OF TOPIC

DIGITAL MONEY AND FINTECH

Revise each lesson in your workbook: the notes, key words and quick checks.
Then take the topic test when Mr Butler sets it.